

Problem

Design an RC circuit to provide a 90° lagging phase shift of the out-put voltage relative to the input voltage. If an ac voltage of 60 V rms is applied, what is the output voltage?

Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

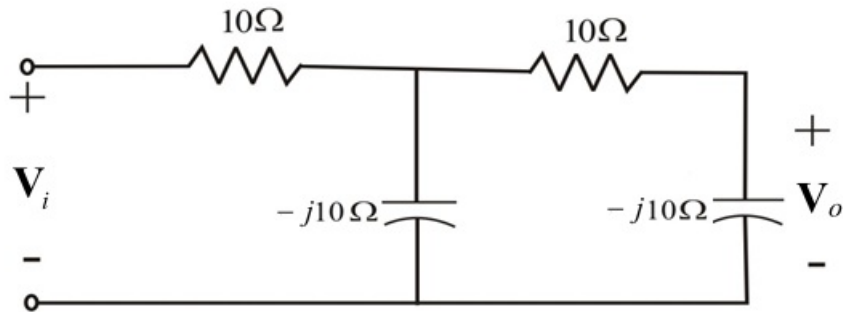


Figure 1

Step 2 of 4

Consider the following:

Z_1 = Impedance of the $-j10\Omega$ inductor in series with 10Ω

Z_2 = impedance of the parallell combination of Z_1 and $-j10\Omega$

Calculate the value of the impedance Z_1 .

$$\begin{aligned} Z_1 &= -j10 + 10 \\ &= (10 - j10)\Omega \end{aligned}$$

Calculate the value of the impedance Z_2 .

$$\begin{aligned} Z_2 &= Z_1 \parallel -j10 \\ &= \frac{(10 - j10)(-j10)}{(10 - j10) - j10} \\ &= (2 - j6)\Omega \end{aligned}$$

Step 3 of 4

Consider V_1 is the voltage across the impedance Z_2 .

Calculate the value of the voltage V_1 .

Write the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= \frac{Z_2}{Z_2 + 10} V_i \\ &= \frac{2 - j6}{2 - j6 + 10} V_i \\ &= \frac{2 - j6}{12 - j6} V_i \end{aligned}$$

$$\begin{aligned} V_1 &= (0.333 - j0.333) V_i \\ &= 0.47 \angle -45^\circ V_i \text{ V} \end{aligned}$$

Therefore, the value of voltage V_1 is $0.47 \angle -45^\circ V_i$ V.

Step 4 of 4

Calculate the value of the output voltage V_o .

$$\begin{aligned} V_o &= \frac{j10}{10 - j10} V_1 \\ &= (-0.5 + j0.5) V_1 \\ &= (0.707 \angle -45^\circ)(0.47 \angle -45^\circ) V_i \\ &= (0.33229 \angle -90^\circ) V_i \text{ V} \end{aligned}$$

Substitute $V_i = 60$ V rms in this expression.

$$\begin{aligned} V_o &= (0.33229 \angle -90^\circ) 60 \text{ V} \\ &= 19.9374 \angle -90^\circ \text{ V} \\ &= 20 \angle -90^\circ \text{ V} \end{aligned}$$

Hence the output voltage V_o magnitude is **20 V** and it lags the input voltage by 90° .

Therefore, the required circuit is as shown in **Figure 1**.